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The weak-form efficiency of the Indian stock market: Fresh Evidence



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The primary goal of the present research initiative is to determine if the Indian stock market follows a random walk or not. The data on eight nifty sectoral indices' daily opening, closing, high and low values are taken from a website. The Augmented Dickey-Fuller test and the Phillips-Perron test is used to assessing the stationarity of the selected eight sectoral indices. The Variance ratio test is used to check for auto-correlation between the returns and the Runs test is performed to examine if the stock market followed a random walk or not. The unit root tests show that the returns from the eight selected sectoral indices are integrated into order 1. According to the variance ratio test future stock prices can be forecasted by prior stock prices in the Indian stock market. The Runs test results indicate the returns are not random throughout the studied time frame.

Introduction

The Efficient Market Hypothesis (EMH) was first proposed in the financial literature by Samuelson (1965) and Fama (1965). According to the EMH, share prices adjust swiftly in reaction to new information, therefore current prices should fully reflect all available information and follow a random walk, which means that subsequent stock price changes (returns) should be distributed independently and equally. EMH described an efficient market as one in which prices always entirely reflect available information.

Three levels of financial market efficiency are defined by EMH-weak, semi-strong, and strong by Fama (1970). Current stock prices, according to the EMH's Weak-Form, represent all securities market information, including past price movements, investment returns, trading volumes, and other market-related information. As a result, any trading strategies to either buy or sell a stock based on prior rates of return or any other historical market data, should yield no benefit. The semi-strong form of the EMH, on the other hand, claims that security prices react immediately to the revelation of all publicly available information. As a result, investors who make decisions based on significant new information after it becomes public should not expect to make above-average risk-adjusted returns from their trades. The EMH's strong form, on the other hand, is the most stringent. It asserts that the stock price accurately reflects all publicly and privately available information. It implies that no investor has exclusive access to key information for price formation. As a result, no investor will ever be able to consistently achieve above-average risk-adjusted returns.

Numerous types of research have been done ever since the efficient market hypothesis notion was created to support or, in some cases, refute the theory. It is practically hard to predict future prices and turn a profit from them when it is clear that prices do, move randomly or drunkenly. On the other hand, if the outcome is the opposite of what was said above, there is a potential to profit simply by looking at a security price's historical behaviour. Because of this, economists and the general investing community are particularly interested in this topic.

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In this study, we focus on the stock market efficiency in its weak form. Prior literature (Chavannavar and Patel, 2016; Jain and Jain, 2013; Asiri and Asiri, 2008) found the Indian stock market to be efficient in its weak form. On the other hand, (Sarkar, 2019; Patel et al., 2018) found the Indian stock market to be weak form inefficient. The literature survey shows a contradictory picture which motivates us to do detailed research in this area. We seek to investigate the Indian stock market's weak-form market efficiency from 2012 to 2021 in this study. The Indian financial market has changed significantly throughout the study.

India's GDP increased from 1.83 trillion USD to 2.66 trillion USD (World Bank, 2022), as did foreign institutional investment and domestic institutional investment. For a rapidly growing economy, the Indian stock market has risen sharply. During our period of study, the Nifty Broad Index has increased by more than three times. Share market participation of Retail Investors of India has increased exponentially as the overall number of Demat accounts increased from 16.8 million to 39.3 million from 2009 to 2019. The internet has become considerably less expensive than previously and all information is reaching everyone in less time. Online trading has been introduced so that anyone can buy and sell shares from anywhere in the world.

We would like to test if there is a weak form of market efficiency or not because the Indian economy has changed significantly in the recent decade. The detailed

investigation is carried out by using different unit root tests, Runs test and variance ratio tests.

Literature Review

The weak form market efficiency is supported and refuted by empirical evidence. Hong (1978), Panas (1990), and Chan et al. (1992) found evidence that stock markets are weak efficient. Hong (1978) used daily data from September 1973 to March 1976 to evaluate market efficiency in Australia, Japan, Hong Kong and Singapore. In comparison to other stock markets, the author discovered that Japanese stocks were more efficient. He went on to say the efficiency of larger stock exchanges was higher. Panas (1990) investigated the weak form market efficiency of the Athens stock market using the autocorrelation function, Kolmogorov Smirnov statistics, runs test, and Von Neumann ratio test. He based his findings on the stock prices of ten companies listed on the Athens stock exchange, concluding that the stock market was inefficient. Chan et al. (1992) investigated the weak-form efficiency in 18 national stock markets both individually and collectively. Their findings supported that the selected stock markets were weak-form efficient. The authors also reported the presence of a contagion effect.

Several studies found the stock markets to be inefficient in their weak form. Previous studies by Worthington and Higgs (2003), Gupta and Basu (2007), Mishra (2009), Srinivasan (2010), Sarkar (2019) and Ahmed (2021) provided evidence relating to the weak form inefficiency of the stock market from European nations,

India, Bangladesh and Saudi, respectively.

Worthington and Higgs (2003) investigated the random walk and weak from market efficiency in European equity markets and found that only Hungary meets the strictest requirements for a random walk-in daily stock returns among emerging countries. While Germany, Ireland, Portugal, Sweden and developed markets, only the United Kingdom meets the strictest random walk criteria. Using monthly closing values of stock market indexes for 14 Asia-Pacific nations, Hamid et al. (2010) conclude that none of the markets perfectly follows the random walk, and hence these markets remained inefficient from January 2004 to December 2009. Rahman et al. (2021) used the DSE General Index (DSEG) and DSE Broad Index (DSEX) to investigate the Random Walk hypothesis of the Bangladesh stock market from 1 January 1993 to 22 November 2001. Throughout the study period, they discovered that the stock market in Bangladesh did not follow a random walk. Li and Li (2016) looked into stock market efficiency and random walk in India, China, Malaysia and Korea, and found mixed results. The Runs test indicated that these markets are random walk, but the variance ratio test rejected the conclusion which implied that all the four markets are inefficient in weak form. Khoj and Akeel (2020) examined the market efficiency of the Saudi Tadawul All Share Index (TASI) and found that it was inefficient from 1 January 2012 to 9 January 2019.

In the Indian context, Gupta and Basu (2007) examined the Sensex and Nifty of India's

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weak form efficiency and concluded that the Indian stock market was weak form inefficient across the studied time. Mishra (2009) examined the Sensex for weak form efficiency during 18 years, from January 1991 to January 2009, and found that BSE was weak form inefficient throughout that time. Elangovan et al. (2022) checked the weak form market efficiency of the Indian stock market using the broad market index. They employed tools such as unit root tests, Runs test and autocorrelation tests. The authors' findings also supported the results of Gupta and Basu (2007) and Mishra (2009).

According to the aforementioned literature review, empirical evidence on weak-form stock market efficiency is inconclusive. Furthermore, most of the studies mentioned above used broad stock market indices to detect the characteristics of weak form efficiency. This study deviates from the previous literature as it has used data on sectoral indices of the Indian stock market.

Data and Methodology

Data sample

The study's purpose is to look into the weak form market efficiency of the Indian stock market. NIFTY Auto, NIFTY Financial Services, NIFTY FMCG, NIFTY IT, NIFTY Media, NIFTY Metal, NIFTY Pharma and NIFTY Realty are the eight sectoral indices examined in the study. The analysis uses daily open, high, low and closing values of chosen sectoral indices from 3 January 2012 to 31 December 2021. For the analysis, we used the average of these four values

rather than the closing prices of individual sectoral indices. All the data have been gathered from the website. The average of four prices was chosen because it removes variations and helps to control volatility.

Methods

The Jarque-Bera test, the Augmented Dickey Fuller test, the Phillips-Perron test, the Runs test, and the Variance ratio test have been used in this study. Khan and Vieito (2012), Bouri et al. (2017) and Angelovska (2018) have used these tools in the past.

The Jarque-Bera (JB) test checks whether a series has the kurtosis and skewness of a normal distribution and is a goodness-of-fit test. The Jarque-Bera test statistic is always positive, and a value that is far from zero indicates that the series does not have a normal distribution. The JB test statistic can be defined with equation 1:

$$JB = \frac{n}{6} [S^2 + \frac{1}{4}(k - 3)^2] \quad (\text{Equation 1})$$

Where, n = number of observations/ degree of freedom; S = sample skewness; K = sample kurtosis.

We have used the JB to test the hypothesis "the series follows normal distribution".

To determine if the sectoral indices' series are stationary, we used the Augmented Dickey Fuller (ADF) unit root test and the Phillips-Perron (PP) unit root test. The null hypothesis is that the series follows a random walk. It is critical to choose the suitable lag duration in both the ADF and PP tests by including enough terms. The Schwarz information criterion (SIC) is used to determine the lag length in this study.

The Runs test is a statistical procedure to decide if a data series is generated from a random process. The test statistic can be computed with the help of the following equation:

$$Z = \frac{R - \bar{R}}{\delta_R} \quad (\text{Equation 2})$$

Where, R = number of observed runs; $\bar{R}$ = number of expected runs and δ_R = standard deviation of number of runs.

Number of expected runs and standard deviation of number of runs can be defined with the help of equation 3 and equation 4:

$$\bar{R} = \frac{2n_1n_2}{n_1 + n_2} + 1 \quad (\text{Equation 3})$$

$$\delta_R = \sqrt{\frac{(2n_1n_2)(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}} \quad (\text{Equation 4})$$

Where, n_1 and n_2 = the number of negative and positive values in the series. We set the null hypothesis of the Runs test as the data series was generated in a random manner.

The variance ratio test helps to determine whether differences in a time series follow a random walk. Lo and MacKinlay (1987) developed the variance ratio test to evaluate the market efficiency hypothesis by determining whether security prices are autocorrelated. If stocks have a high degree of correlation, historical prices can be used to forecast future prices. This situation violates the weak-form market efficiency hypothesis. Thus, the present study sets three hypotheses:

H_01 = the indices flows normal distribution.

H_02 = the indices has unit root.

H_03 = the indices are generated in a random manner.

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Findings

Table 1 Summary Statistics

	NIFTY Auto	NIFTY Financial Services	NIFTY FMCG	NIFTY IT	NIFTY Media	NIFTY Metal	NIFTY Pharma	NIFTY Realty
Mean	4.605642	4.605830	4.605696	4.605912	4.605447	4.605491	4.605628	4.605562
Maximum	4.678504	4.695831	4.663530	4.671236	4.707923	4.681303	4.697768	4.692914
Minimum	4.504085	4.494494	4.533239	4.514328	4.513863	4.515734	4.534105	4.468306
Skewness	-0.597523	-0.314476	-0.098561	-0.791404	-0.201148	-0.278971	0.050117	-0.542413
Kurtosis	10.94469	11.99347	11.01797	12.18418	8.740368	5.270998	9.124951	6.769017
Jarque-Bera	6656.334	8381.807	6633.693	8956.857	3414.846	563.9631	3869.773	1586.304
Probability	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
Observations	2475	2475	2475	2475	2475	2475	2475	2475

Source: Authors' Calculation

Table 1 reports the summary statistics of the indices' returns. From the table, it can be observed that all the indices except the NIFTY Pharma index are negatively skewed. Analysing the kurtosis values indicates that all the return

series follow the platykurtic distribution.

Jarque-Bera test examines the variables and whether the series follow a normal distribution. Thus, a p-value of less than 0.05 will indicate the series is not normally distributed at the 5%

significance level. Analysing the Jarque-Bera statistics reveals that none of the return series are normally distributed. The results of the skewness, kurtosis and Jarque-Bera test suggest that the returns of the selected sectoral indices are distributed asymmetrically.

Table 2 Unit Root Tests

Variables	ADF		PP	
	Constant (at level)	Constant + Trend (at level)	Constant (at level)	Constant + Trend (at level)
NIFTY Auto	-31.7076*	-31.7271*	-36.5958*	-36.5864*
NIFTY Financial Services	-11.3633*	-11.3590*	-38.4550*	-38.4474*
NIFTY FMCG	-13.9775*	-14.0172*	-37.5352*	-37.5428*
NIFTY IT	-26.3877*	-26.4283*	-38.1131*	-38.1168*
NIFTY Media	-12.6184*	-12.6441*	-35.0492*	-35.0567*
NIFTY Metal	-14.3538*	-14.3995*	-36.6527*	-36.5838*
NIFTY Pharma	-24.7461*	-24.7501*	-35.2525*	-35.2497*
NIFTY Realty	-25.3372*	-25.3453*	-33.9717*	-34.3440*

Source: Authors' Calculation

Note: *represents significance at the 1% level

The output of the ADF test and PP test are summarised in Table 2. We have performed the unit root tests with both constant and constant + trend in the equation. If a series is

non-stationary then the mean, variance and co-variance of the returns are not constant over time. However, a stationary series' mean, variance and co-variance do not change over

time. The output of the ADF test reveals that the selected NIFTY sectoral indices are stationary at level series (the test statistic is significant at the 1 per cent level indicating

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the computed test statistic is more negative than the critical values). The output of the PP test supports the results of the

ADF test and shows that the sectoral indices are integrated into order 1. The presence of the stationarity property in

the sectoral indices suggests that the Indian stock market is inefficient in its weak form.

Table 3 Runs Test

	Mean	n_0	n_1	n_0+n_1	Number of Runs	Expected Runs	Z statistic
Nifty Auto Index	4.605642344	1199	1276	2475	955	1237.302	-11.362*
Nifty financial services index	4.605829977	1215	1260	2475	953	1238.090	-11.467*
Nifty FMCG Index	4.605696286	1221	1254	2475	973	1238.28	-10.668*
Nifty IT Index	4.605911804	1218	1257	2475	969	1238.192	-10.826*
Nifty Media Index	4.605447334	1220	1255	2475	947	1238.252	-11.713*
Nifty Metal Index	4.605491194	1223	1252	2475	939	1238.3301	-12.037*
Nifty Pharma Index	4.605627868	1196	1279	2475	937	1237.108	-12.080*
Nifty Realty Index	4.605561983	1166	1309	2475	895	1234.368	-13.691*

Source: Authors' Computation

n_0 = Cases < Mean value, n_1 = Cases $\geq$ Mean value and $n_0 + n_1$ = Total number of cases.

*represents statistically significant at the 1% level

The returns of the selected Nifty sectoral Indices are subjected to the Runs test. Table 3 summarises the results of the Runs test based on the Mean. The actual number of runs is lower than the expected number of runs, and the difference is

significant across all sectoral indices (computed z-statistic is statistically significant at the 1 percent level). Thus, at the 1% significance level, the null hypothesis of randomness is rejected, indicating that these sector returns from January 2012 to December 2021 are

not random. As a result, the Alternative Hypothesis "No randomness in Nifty sectoral Indices returns" can be accepted. The output of the Runs test shows that the Indian stock market is inefficient in its weak form.

Table 4 Variance Ratio Test of NIFTY Auto index returns

	Period	Var. Ratio	Std. Error	z-Statistic	Probability
NIFTY Auto	2	0.694805	0.050407	-6.054591	0.0000
	4	0.348964	0.087459	-7.443935	0.0000
	8	0.167209	0.127321	-6.540863	0.0000
	16	0.087152	0.179133	-5.095925	0.0000
NIFTY Financial services	2	0.683415	0.053586	-5.908031	0.0000
	4	0.328281	0.092384	-7.270909	0.0000
	8	0.156902	0.136190	-6.190614	0.0000
	16	0.084363	0.192835	-4.748295	0.0000
NIFTY FMCG	2	0.674431	0.066712	-4.880197	0.0000
	4	0.333729	0.110485	-6.030425	0.0000
	8	0.172468	0.149749	-5.526113	0.0000
	16	0.089940	0.198970	-4.573860	0.0000

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NIFTY IT	2	0.674004	0.056075	-5.813550	0.0000
	4	0.333777	0.093216	-7.147091	0.0000
	8	0.170296	0.125491	-6.611665	0.0000
	16	0.083725	0.166586	-5.500301	0.0000
NIFTY Media	2	0.674004	0.056075	-5.813550	0.0000
	4	0.333777	0.093216	-7.147091	0.0000
	8	0.170296	0.125491	-6.611665	0.0000
	16	0.083725	0.166586	-5.500301	0.0000
NIFTY Metal	2	0.741403	0.039309	-6.578596	0.0000
	4	0.373448	0.069291	-9.042347	0.0000
	8	0.183463	0.101275	-8.062605	0.0000
	16	0.094028	0.140654	-6.441126	0.0000
NIFTY Pharma	2	0.697493	0.038135	-7.932508	0.0000
	4	0.351637	0.065310	-9.927541	0.0000
	8	0.168811	0.092991	-8.938344	0.0000
	16	0.089111	0.130718	-6.968339	0.0000
NIFTY Realty	2	0.712179	0.045345	-6.347315	0.0000
	4	0.367637	0.078010	-8.106157	0.0000
	8	0.182512	0.114054	-7.167568	0.0000
	16	0.096686	0.159495	-5.663569	0.0000

Source: Authors' Computation

The results of the variance ratio test are presented in Table 4. It can be observed that the computed z-statistics for period 2, period 4, period 8 and period 16 for all NIFTY sectoral indices are negative and statistically significant at the 1 per cent level. The significant z-statistics show that the computed z-statistics are more than the critical values. Thus, the null hypothesis of the random walk can be rejected. The results show that the returns of the NIFTY sectoral indices are correlated. This further validates that the Indian stock market is weak form inefficient which supports the results of unit root tests and Runs tests.

Discussion and Conclusion

The study's main goal is to see if the Indian stock market contains characteristics of a

weakly efficient market. In addition, the distribution of sectoral returns is studied to see if there is any asymmetry in the returns. We examined the years



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2012-2021, which were marked by economic changes and substantial improvements in the Indian stock market. The study's findings reveal that the skewness and kurtosis of the selected sectoral returns deviate from the skewness and kurtosis of a normal distribution. The findings reveal that the returns are asymmetrical.

As a result, the study's null hypothesis H_01 is rejected.

The returns of the selected sectoral indexes are stationary, according to the ADF and PP unit root tests. The unit root test results reveal that the future returns of the sectoral indices can be predicted based on past returns. Thus, the null hypothesis H_02 is rejected. The output of the Runs test and variance ratio test supports the proposition of weak form inefficiency. Thus, H_03 of the study can also be rejected. The findings of the study are consistent with the previous study of Gupta and Basu (2007), Mishra (2009) and Elangovan et al. (2022). The overall results show that there is asymmetry as well as the absence of randomness in the sectoral returns indicating that the Indian stock market is weak form inefficient. We may offer a few plausible reasons why the Indian stock market exhibits weak-form inefficiency. Behavioural finance asserts that investors are guided more by psychology rather than rationality and efficiency. Thus, the Indian stock market's weak-form inefficiency is



Technical analysis, which examines previous stock price movements and volume patterns to predict future price patterns, may be a significant technique for Indian stock market investors to create consistent excess returns.



due to investor emotions, cognitive errors, market shocks illiquidity and speculative economic bubbles.

The findings of this study provide useful investment techniques for Indian stock market investors. Technical analysis, which examines previous stock price movements and volume patterns to predict

future price patterns, may be a significant technique for Indian stock market investors to create consistent excess returns.

Future research endeavours can divide the total period time into sub-periods to investigate the time-varying nature of the weak form efficiency of the Indian stock market. Furthermore, researchers can employ advanced techniques like wavelet analysis in their research work. The use of high frequency data on stock price movement can also be used for testing the weakform market efficiency of the Indian stock market.

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